

Activity 4 — Pareto Analysis to Target the Vital Few

Nexa Logistics — 2,000 failed deliveries a month

Topic	Topic 1
Objective	A1, A3 · K2 — Prioritise causes by contribution and identify key implications.
Duration	40 minutes
Grouping	Teams of 3-5
Tools	Pareto Chart ed-tool, ChatGPT / Copilot / Gemini
Ed-tool	Pareto Chart Tool — https://alfredang.github.io/paretochart/

The Scenario

Nexa Logistics runs last-mile delivery for e-commerce clients across Singapore. Of 40,000 monthly deliveries, 2,000 fail on first attempt — a 5% failure rate against a contractual SLA of 2%. Every failed delivery costs S\$8.50 to re-attempt: S\$17,000 a month, and two clients have invoked SLA penalty clauses.

The Operations Director has a list of eleven failure reasons and a budget that will fund roughly two initiatives this quarter. His instinct is to start with 'Address incorrect' because it generates the angriest client emails.

Your team must decide where the two initiatives should go — using the data, not the noise.

The Evidence

Nexa Logistics — first-attempt delivery failures by reason, last month

Failure reason	Count	Cost impact (\$)
Recipient not at home (no delivery window given)	742	6307
Driver could not access condo / office lobby	486	4131
Address incomplete or incorrect	231	1964
Parcel not loaded (sorting error)	188	1598
Recipient refused (wrong item expected)	121	1029
Vehicle breakdown / route disruption	94	799
Weather-related suspension	61	519
Recipient contact number invalid	38	323
Parcel damaged in transit	24	204
Restricted delivery hours at destination	20	170
Other / unclassified	15	128

Discussion Questions

1. Build the Pareto chart. How many of the eleven reasons account for roughly 80% of failures? What does that ratio mean for how the Operations Director should spend his budget?
2. The Director wants to start with 'Address incorrect' because it generates the angriest emails. What does the data say, and how would you make that argument to him without dismissing his concern?
3. Calculate: if the top two reasons drop by 60%, what is the new failure rate? Does that meet the 2% SLA? Show your arithmetic.
4. The top two reasons — 'not at home' and 'lobby access' — look like customer problems. Reframe each as something Nexa controls. What changes about who owns the fix?
5. Pareto ranks by FREQUENCY. Name a situation in your own workplace where the most frequent problem is not the most important one. What would you rank by instead?

Step-by-Step

6. Read the Nexa Logistics scenario and the failure reason table.
7. Open the Pareto ed-tool at <https://alfredang.github.io/paretochart/>
8. Enter each failure reason and its count. The tool sorts descending and computes the cumulative line.
9. Read off the cumulative percentage and mark where the line crosses 80%.
10. Count how many reasons sit to the left of that line — these are your vital few.
11. Calculate by hand: a 60% reduction on the top two gives what new failure rate? Compare against the 2% SLA.
12. Run the GenAI Pareto prompt from the slide and check the AI's arithmetic against your own. Note any error.
13. Restate the top two reasons so that Nexa — not the customer — owns the fix.
14. Decide where the two funded initiatives should go and prepare a one-minute case for the Operations Director.
15. Present, including the arithmetic showing whether the SLA is reachable.

The GenAI Prompt

Act as an operations analyst. Perform a Pareto (80/20) analysis on the delivery failure data below for a Singapore last-mile logistics company.

Total deliveries: 40,000/month. Failures: 2,000 (5%). SLA target: 2%. Re-attempt cost: S\$8.50 each.

Data (reason, count):

Recipient not at home 742; Lobby access denied 486; Address incorrect 231; Parcel not loaded 188; Recipient refused 121; Vehicle breakdown 94; Weather 61; Invalid contact 38; Damaged 24; Restricted hours 20; Other 15.

Produce:

1. A table with count, percentage, and CUMULATIVE percentage, sorted descending
2. The cut-off line – which reasons make up the vital few (roughly 80%)
3. The monthly S\$ recovery if the top 2 reasons were reduced by 60%
4. Whether hitting the 2% SLA is achievable by fixing the vital few alone – show the arithmetic
5. One caution about what this Pareto analysis does NOT tell me

Trainer Debrief

Expected answers and the points to draw out. Trainer reference.

EXPECTED ANALYSIS:

Not at home 742 (37.1%, cum 37.1%)
Lobby access 486 (24.3%, cum 61.4%)
Address incorrect 231 (11.6%, cum 73.0%)
Parcel not loaded 188 (9.4%, cum 82.4%) <- the 80% line falls here
The remaining 7 reasons together account for under 18%.

So FOUR of eleven reasons carry 82% of the failures — and the top TWO alone carry 61%.

ARITHMETIC: a 60% reduction on the top two removes $0.6 \times 1,228 = 737$ failures, leaving 1,263 of 40,000 = 3.2%. That is a large win but it does NOT reach the 2% SLA (which needs failures under 800). Reaching SLA requires the top two AND a meaningful dent in reasons 3 and 4. Learners who claim the SLA is met have not done the arithmetic — make them show it.

KEY TEACHING POINTS:

1. 'Address incorrect' is only 11.6% — third place. It is loud, not large. The Director is responding to escalation volume, which correlates with client temperament, not failure frequency. The Pareto chart is how you have that conversation with evidence rather than opinion. This is the transferable political skill in this activity.
2. REFRAMING OWNERSHIP is the deeper lesson: 'recipient not at home' is only a customer problem if you never offered a delivery window. 'Lobby access denied' is only a customer problem if you never negotiated building access protocols. Both become Nexa-controllable the moment you restate them — and 61% of the problem changes owner.
3. LIMITATION: Pareto ranks by frequency, and frequency is not severity. A single safety-critical or regulatory failure can outrank 700 inconveniences. Always ask what the right ranking dimension is — count, cost, risk or customer harm. Here, cost tracks count closely (uniform S\$8.50), which is why the two rankings agree; that is not always true.
4. Pareto tells you WHERE to look, never WHY. The 742 'not at home' failures still need a 5 Whys. The tools chain: Pareto for priority, then 5 Whys for cause.

Self-Check — Is Your Output Finished?

Your analysis is sound when: the cumulative column reaches 100%; you can name the vital few and the exact cumulative percentage at your cut-off; you have shown the arithmetic on whether the 2% SLA is reachable (it is not, on the top two alone); and you can state in one sentence what Pareto does NOT tell you. Verify the AI's percentages against the tool's — if they disagree, the tool is right and you have just learned something about trusting AI arithmetic.